Grade 12 Practice

Standard S.ID.3

Name: _____ Date: _____

Computation

1. Adding an outlier affects which measure the most? Write mean or median.

mean

2. Does the median resist outliers? Write yes or no.

yes

3. Data are 1, 2, 3, 4. If 100 is added, does the mean increase? Write yes or no.

yes

4. Does an outlier affect the range? Write yes or no.

yes

Word problems

5. Salaries include one very high CEO salary. Which center should be reported to show a typical salary? Write mean or median.

Show your work and answer.

Answer: median

6. Data are 2, 3, 4, 5, and 50 is added. What is the new range?

Show your work and answer.

Answer: 48

Grade 12 Practice

Standard S.ID.4

Name: _____ Date: _____

Computation

1. A normal distribution has mean 50 and standard deviation 10. What is the z-score of 60? _____ **1**

2. What is the z-score of 40 in the same distribution? _____ **-1**

3. In a normal distribution, about what percent of data falls within one standard deviation of the mean? _____ **68%**

4. What is the z-score of the mean? _____ **0**

Word problems

5. A test has mean 70 and standard deviation 5. What is the z-score of a score of 80?

Show your work and answer.

Answer: **2**

6. Heights are normal with mean 65 inches and standard deviation 3 inches. About what percent lie within one standard deviation?

Show your work and answer.

Answer: **68%**

Grade 12 Practice

Standard S.IC.1

Name: _____ Date: _____

Computation

1. A census surveys every member of a population. Write yes or no: is this a sample?

no

2. A survey asks 100 students from a school of 1,000. Is this a sample? Write yes or no.

yes

3. Is a sample a subset of a population? Write yes or no.

yes

4. A parameter describes a population. What does a statistic describe? Write your answer.

sample

Word problems

5. A poll of 500 voters is taken in a city of 80,000. What is the sample size?

Show your work and answer.

Answer: **500 voters**

6. The average lifetime of all bulbs a factory makes is a _____. Write parameter or statistic.

Show your work and answer.

Answer: **parameter**

Grade 12 Practice

Standard S.IC.2

Name: _____ Date: _____

Computation

1. A fair coin is modeled with 2 equal outcomes. Is this a valid model for a 50% chance? Write yes or no.

yes

2. A 10% chance is modeled with 10 equal outcomes, where 1 counts as success. Is this valid? Write yes or no.

yes

3. A simulation results are compared to the theoretical probability. Is this a valid check? Write yes or no.

yes

4. A 1/4 chance is modeled with 4 equal outcomes, where 1 counts as success. Is this valid? Write yes or no.

yes

Word problems

5. A spinner has 3 equal sections. Is it a valid model for a 1/3 chance? Write yes or no.

Show your work and answer.

Answer: **yes**

6. A player makes 70% of shots. Is a model with 10 equal outcomes, where 7 count as success, valid? Write yes or no.

Show your work and answer.

Answer: **yes**

Grade 12 Practice

Standard S.IC.3

Name: _____ Date: _____

Computation

1. Is a random sample unbiased? Write yes or no.

yes

2. Surveying only students who ride the bus is biased. Write yes or no.

yes

3. Is a convenience sample random? Write yes or no.

no

4. Does a stratified sample use groups? Write yes or no.

yes

Word problems

5. A teacher picks every 10th name from an alphabetical list. What sampling method is this?

Show your work and answer.

Answer: systematic

6. A survey picks 20 students from each grade level. What sampling method is this?

Show your work and answer.

Answer: stratified

Grade 12 Practice

Standard S.IC.4

Name: _____ Date: _____

Computation

1. A sample of size 100 has a margin of error of about how many percent?

_____

10%

2. Does the margin of error decrease when sample size increases? Write yes or no.

yes

3. A sample of size 400 has a margin of error of about how many percent?

_____

5%

4. A confidence interval is written as estimate ± _____.

margin of error

Word problems

5. A poll shows 60% support with a margin of error of 3%. What is the confidence interval?

Show your work and answer.

Answer: **57% to 63%**

6. A sample of 1,000 has a margin of error of about how many percent?

Show your work and answer.

Answer: **about 3%**

Name: _____ Date: _____

Computation

1. Does random assignment help show cause? Write yes or no.

yes

2. What is the purpose of a control group? Write your answer in one word.

comparison

3. If a difference is larger than chance would explain, is it significant? Write yes or no.

yes

4. What is the purpose of a placebo? Write blinding or placebo effect control.

blinding

Word problems

5. A treatment group improves by 8 points and a control group improves by 2 points. What is the difference?

Show your work and answer.

Answer: 6 points

6. Without random assignment, can you prove causation? Write yes or no.

Show your work and answer.

Answer: no

Name: _____ Date: _____

Computation

1. If a sample is biased, are the conclusions limited? Write yes or no. **yes**

2. Does correlation prove causation? Write yes or no. **no**

3. Is a sample of 10 people always representative of a city? Write yes or no. **no**

4. Is a larger random sample generally more reliable? Write yes or no. **yes**

Word problems

5. A report says a drug works based on 20 people with no control group. Is it strong evidence? Write yes or no.

Show your work and answer.

Answer: **no**

6. A survey was only done online. Can it be used to describe all adults? Write yes or no.

Show your work and answer.

Answer: **no**

Grade 12 Practice

Standard S.CP.1

Name: _____ Date: _____

Computation

1. How many outcomes are in the sample space for rolling one die? _____ **6**

2. How many outcomes are in the event 'even' when rolling a die? _____ **3**

3. If $P(A) = 0.3$, what is the probability of the complement of A? _____ **0.7**

4. How many outcomes are in the sample space for flipping two coins? _____ **4**

Word problems

5. How many cards in a standard deck are red? (Event: red card.)

Show your work and answer.

Answer: 26

6. When one die is rolled, the event 'roll greater than 4' has how many outcomes?

Show your work and answer.

Answer: 2

Name: _____ Date: _____

Computation

1. P(A) = 0.5, P(B) = 0.4, and P(A and B) = 0.2. Are A and B independent? Write yes or no.

yes

2. P(A) = 0.3, P(B) = 0.6, and P(A and B) = 0.1. Are A and B independent? Write yes or no.

no

3. Are two coin flips independent? Write yes or no.

yes

4. Are two draws without replacement independent? Write yes or no.

no

Word problems

5. Whether it rains and whether a sunny day occurs are independent? Write yes or no.

Show your work and answer.

Answer: **no**

6. Two dice are rolled. Are the results independent? Write yes or no.

Show your work and answer.

Answer: **yes**

Grade 12 Practice

Standard S.CP.3

Name: _____ Date: _____

Computation

1. P(A and B) = 0.2 and P(B) = 0.5. What is P(A | B)? _____

0.4

2. P(A and B) = 0.09 and P(A | B) = 0.3. What is P(B)? _____

0.3

3. If A and B are independent, P(A | B) equals _____.

P(A)

4. Write the formula for P(A | B).

P(A and B) / P(B)

Word problems

5. Of 15 students who play sports, 10 are girls. What is P(girl | plays sports)? Write as a fraction.

Show your work and answer.

Answer: 2/3

6. P(B) = 0.25 and P(A and B) = 0.05. What is P(A | B)?

Show your work and answer.

Answer: 0.2

Grade 12 Practice

Standard S.CP.4

Name: _____ Date: _____

Computation

1. A survey of 100 people has Group A with 30 yes and 20 no, and Group B with 10 yes and 40 no. How many people were surveyed in all? _____

100

2. What is P(yes | Group A)? Write as a fraction.

3/5

3. What is P(yes | Group B)? Write as a fraction.

1/5

4. What is P(Group A)? Write as a fraction.

1/2

Word problems

5. Which group is more likely to say yes? Write A or B.

Show your work and answer.

Answer: A

6. What is P(Group A and no)? Write as a fraction.

Show your work and answer.

Answer: 1/5

Grade 12 Practice

Standard S.CP.5

Name: _____ Date: _____

Computation

1. $P(A \mid B)$ means the probability of A given that _____ happened.

B

2. Two independent events both occur. Their probabilities are multiplied. Write yes or no.

yes

3. $P(A) = 0.5$ and $P(B \mid A) = 0.2$. What is $P(A \text{ and } B)$? _____

0.1

4. $P(B \mid A) = 0.2$ and $P(B) = 0.2$. Are A and B independent? Write yes or no.

yes

Word problems

5. Events A and B are independent. $P(A) = 0.3$ and $P(B) = 0.4$. What is $P(A \text{ and } B)$?

Show your work and answer.

Answer: 0.12

6. Two components each work with probability 0.9. What is the probability both work?

Show your work and answer.

Answer: 0.81

Grade 12 Practice

Standard S.CP.6

Name: _____ Date: _____

Computation

1. Of 200 cars, 50 have GPS and 40 of those have a radio. What is P(radio | GPS)?

0.8

2. What is P(GPS) out of 200 cars? Write as a fraction.

1/4

3. What is P(GPS and radio) out of 200 cars? Write as a fraction.

1/5

4. Radio is in 80 of the 200 cars. Are GPS and radio independent? Write yes or no.

no

Word problems

5. A bag has 5 red and 3 blue marbles. Two are drawn without replacement. What is P(red then blue)? Write as a fraction.

Show your work and answer.

Answer: **15/56**

6. The same bag is used, but the first marble is replaced. What is P(red then blue)? Write as a fraction.

Show your work and answer.

Answer: **15/64**

Grade 12 Practice

Standard S.CP.7

Name: _____ Date: _____

Computation

1. $P(A) = 0.4$, $P(B \mid A) = 0.5$, and $P(B) = 0.25$. What is $P(A \mid B)$?

0.8

2. $P(A) = 0.2$ and $P(B \mid A) = 0.5$. What is $P(A \text{ and } B)$?

0.1

3. If $P(B \mid A) = P(B)$, are the events independent? Write yes or no.

yes

4. $P(A) = 0.6$ and $P(B \mid A) = 0.1$. What is $P(A \text{ and } B)$?

0.06

Word problems

5. $P(A) = 0.3$, $P(B \mid A) = 0.4$, and $P(B) = 0.2$. What is $P(A \mid B)$?

Show your work and answer.

Answer: 0.6

6. A spam filter flags 90% of spam. 20% of email is spam, and 25% of email is flagged. What is $P(\text{spam} \mid \text{flagged})$?

Show your work and answer.

Answer: 0.72

Grade 12 Practice

Standard S.MD.1

Name: _____ Date: _____

Computation

1. A game pays $10 with probability 0.1 and $0 otherwise. What is the expected value?

$1

2. A game pays $5 with probability 1/2 and costs $1 otherwise. What is the expected value?

2

3. What is the expected value of a fair game? _____

0

4. What is the expected value of rolling a fair die and winning the number shown? _____

3.5

Word problems

5. A lottery ticket costs $2 and pays $100 with probability 1/100. What is the expected net value?

Show your work and answer.

Answer: -$1

6. A random number from 1 to 4 pays that many dollars. What is the expected payout?

Show your work and answer.

Answer: $2.50

Grade 12 Practice

Standard S.MD.2

Name: _____ Date: _____

Computation

1. A random variable X is 1, 2, or 3, each with probability 1/3. What is E(X)? **2**

2. X is 0 with probability 0.5 and 4 with probability 0.5. What is E(X)? **2**

3. X is 10 with probability 0.2 and 0 with probability 0.8. What is E(X)? **2**

4. X is 3 with probability 0.25 and 1 with probability 0.75. What is E(X)? **1.5**

Word problems

5. A store sells 0, 1, or 2 items with probabilities 0.5, 0.3, and 0.2. What is the expected number of items?

Show your work and answer.

Answer: 0.7 items

6. A game pays $20 with probability 0.1 and costs $2 otherwise. What is the expected value?

Show your work and answer.

Answer: $0.20

Name: _____ Date: _____

Computation

1. Do the probabilities of a distribution have to sum to 1? Write yes or no. **yes**

2. Are probabilities 0.2, 0.3, and 0.5 a valid distribution? Write yes or no. **yes**

3. Are probabilities 0.4, 0.4, and 0.4 a valid distribution? Write yes or no. **no**

4. If $P(0) = 0.2$ and $P(1) = 0.3$, what is $P(X \leq 1)$? _____ **0.5**

Word problems

5. A fair die is modeled with $P(X = 1)$ through $P(X = 6)$. What do the six probabilities sum to?

Show your work and answer.

Answer: 1

6. Probabilities for 0, 1, and 2 are 0.5, 0.3, and x. What is x?

Show your work and answer.

Answer: 0.2

Grade 12 Practice

Standard S.MD.4

Name: _____ Date: _____

Computation

1. When choosing between two options, should you pick the one with higher expected value? Write yes or no.

yes

2. Option A has an expected value of 10 and option B has 8. Which should you choose? Write A or B.

A

3. A decision is based only on expected value. Is this always the best choice? Write yes or no.

no

4. Expected value 0 means the game is what? Write fair or unfair.

fair

Word problems

5. Option A has an expected profit of $500. Option B has $450. Which option should a company pick?

Show your work and answer.

Answer: A

6. Two games have expected values of -$0.50 and $0. Which game should you choose?

Show your work and answer.

Answer: the $0 game

Grade 12 Practice

Standard S.MD.6

Name: _____ Date: _____

Computation

1. Is a game with expected value 0 fair? Write yes or no. **yes**

2. A game wins $4 with probability 0.2 and loses $1 with probability 0.8. Is it fair? Write yes or no. **yes**

3. A spinner has 4 equal sections, each with a prize. Is the spinner fair? Write yes or no. **yes**

4. A game has expected value -0.25. Is it in your favor? Write yes or no. **no**

Word problems

5. A dice game pays $6 on a 6 and costs $1 otherwise, with probabilities 1/6 and 5/6. Is it fair? Write yes or no.

Show your work and answer.

Answer: no

6. You pay $5 to enter and win $10 with probability 0.5. Is it fair? Write yes or no.

Show your work and answer.

Answer: yes